CCZG506 | API-DRIVEN CLOUD NATIVE SOLUTIONS

API-Driven Chest X-Ray Analysis and Explanation Assistant

Assignment II - Integrated Computer Vision, Grounded NLP, FastAPI, Streamlit and LLMOps

DOMAIN	AI CATEGORIES	GROUP
Healthcare / Medical Imaging	**Computer Vision + NLP**	**Group 89**

Educational-use boundary: This prototype supports research and educational demonstration. It is not a diagnostic system and does not replace review by a qualified healthcare professional.

BITS Pilani Work Integrated Learning Programmes

0. Group Details and Contribution

The contribution allocation is balanced across five core workstreams. Each member is assigned a distinct Assignment II responsibility while the integrated deliverable remains a collective group outcome.

#	BITS ID	Name	Assignment II contribution	Weight
1	2024ac05968@wilp.bits-pilani.ac.in	ADURTI SAI VENKATESH	ChestMNIST dataset acquisition and integrity validation, multilabel exploratory analysis, preprocessing design, class-imbalance analysis, and data-engineering evidence.	20%
2	2024ac05284@wilp.bits-pilani.ac.in	ANKITA TOMAR	ResNet-18 multilabel model development, training-strategy review, checkpoint analysis, finding-specific threshold calibration, and model-development evidence.	20%
3	2024ac05653@wilp.bits-pilani.ac.in	D MALLIKARJUNA REDDY	FastAPI contract and service integration, Streamlit HTTP-client workflow, end-to-end interface orchestration, packaging coordination, and deployment validation.	20%
4	2024ad05005@wilp.bits-pilani.ac.in	KSHIRSAGAR RUTUJA VILAS	Isolated test evaluation, multilabel error analysis, Grad-CAM explainability validation, representative evidence review, and interpretation of model limitations.	20%
5	2024aa05099@wilp.bits-pilani.ac.in	MISHRA SUDHAKAR RAMNAWAL LALIT	Grounded-language task design, prompt and guardrail review, language evaluation, LLMOps and MLflow evidence, documentation checks, and demonstration support.	20%

Contribution total: 5 members x 20% = 100% of the Assignment II work allocation.

Table of Contents

1. Executive Summary

This project designs and implements an interactive, API-driven educational assistant for chest X-ray analysis. It combines a fine-tuned multilabel computer-vision model, finding-specific Grad-CAM attribution, a fine-tuned grounded language model, deterministic safety guardrails, a versioned FastAPI service, and a Streamlit demonstration interface. The unifying principle is that every language response must be derived from structured computer-vision evidence rather than independent image interpretation.

The computer-vision component fine-tunes ImageNet-pretrained ResNet-18 on ChestMNIST at 224 x 224 resolution. It produces fourteen independent probabilities and applies frozen, finding-specific thresholds selected on validation data. The isolated test set contains 22,433 images. Final test performance includes macro ROC-AUC 0.8175, micro ROC-AUC 0.8226, macro F1 0.2816, micro F1 0.3391 and Hamming loss 0.0988.

The natural-language component performs full sequence-to-sequence fine-tuning of FLAN-T5-small on 4,800 grounded records spanning structured reports, plain-language explanations, question answering and educational follow-up. Held-out evaluation on 600 records produced ROUGE-L F1 0.8489 and smoothed BLEU-4 0.7393. A deterministic guardrail processed all generations and achieved complete post-guardrail contract compliance, including a zero unsupported-finding rate.

Result: The final readiness gate passed all 20 checks, with 49 checksum-registered artifacts, 23 passing API tests, 5 passing HTTP-client tests, 5 passing UI state tests, and an operational live workflow.

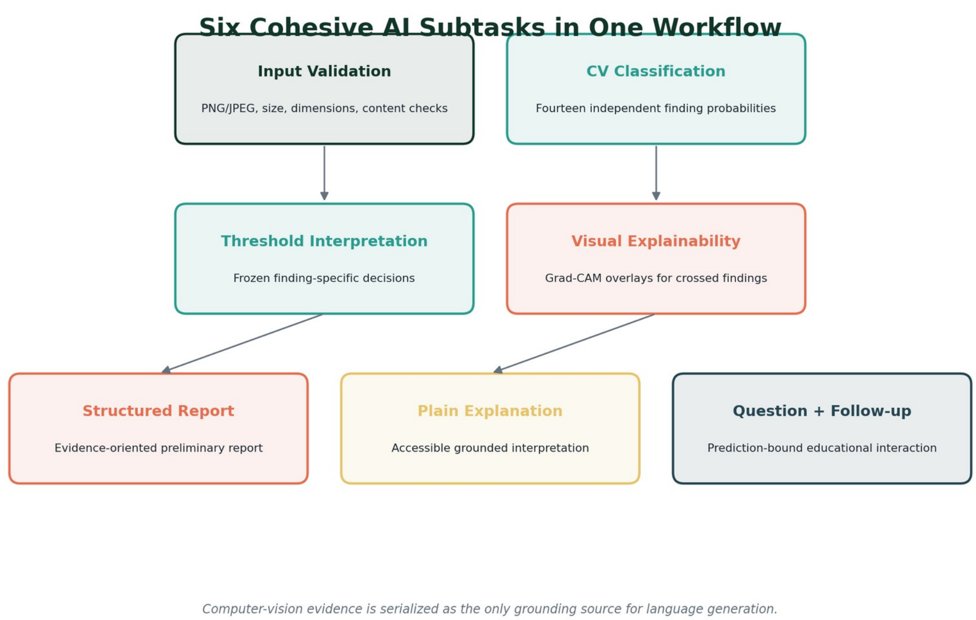

Figure 1. The six cohesive AI subtasks and their shared evidence flow. Source: Project design synthesized from Notebooks 2-8.

2. Assignment Alignment and Unified Objective

The selected domain is healthcare and medical imaging. The chosen AI categories are Computer Vision and Natural Language Processing. Six cohesive subtasks are delivered, exceeding the minimum requirement of five. They are integrated through a single evidence contract and exposed as an interactive API-based application.

Category	Subtask	Model/service	Integrated output
Computer Vision	Multilabel image classification	ResNet-18	Fourteen finding probabilities and frozen decisions
Computer Vision	Visual explainability	LayerGradCam	Finding-specific overlays and attribution heatmaps
NLP	Structured report generation	FLAN-T5-small	Evidence-oriented preliminary report
NLP	Plain-language explanation	FLAN-T5-small	Accessible grounded explanation
NLP	Grounded question answering	FLAN-T5-small + guardrail	Prediction-bound answer
NLP	Educational follow-up	FLAN-T5-small + guardrail	Controlled next-step guidance

Cohesion is enforced technically. The vision service emits a strict finding-evidence schema containing label identity, probability, calibrated threshold, threshold decision, confidence category and approved description. The language serializer consumes this schema and the guardrail prohibits unsupported finding names, ungrounded numeric claims, diagnostic language and omission of required limitations.

Ground truth: Ground truth refers to the reference labels supplied by ChestMNIST for training and evaluation. It is not the application's prediction and is not equivalent to a new clinical diagnosis.

3. Architecture and End-to-End Workflow

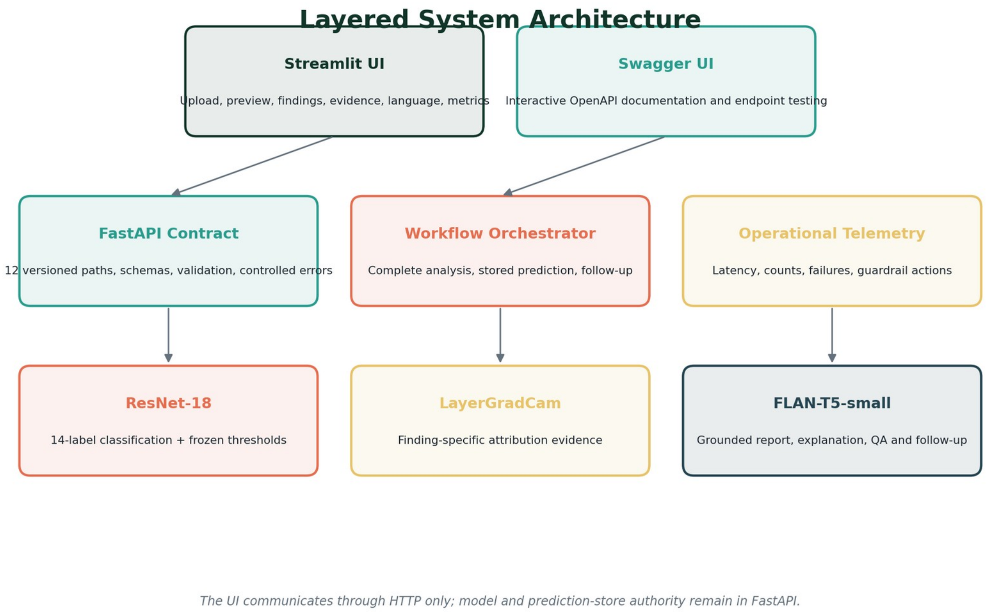

Figure 2. Layered architecture separating interface, API orchestration, model services and persisted evidence.

The Streamlit interface is intentionally thin. It validates only the immediate user-experience contract, such as accepted file extensions and preview availability. Authoritative validation, model loading, inference, threshold application, Grad-CAM generation, language generation, guardrail auditing and prediction persistence occur in FastAPI services.

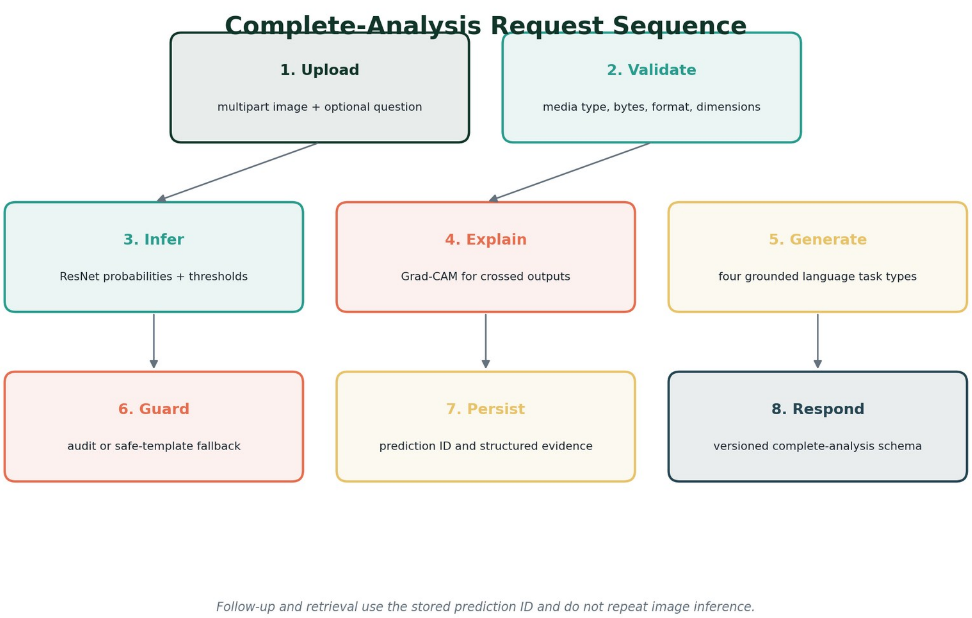

Figure 3. Complete-analysis sequence and prediction reuse.

A single POST request to /api/v1/analyze-complete triggers the full workflow. The response contains request and prediction identifiers, image metadata, all fourteen findings, explainability metadata, optional visual evidence, multiple language outputs, versions, warnings and latency. Follow-up endpoints receive only the prediction identifier and question; the client cannot supply or modify grounding evidence.

4. Development Method and Notebook Traceability

Figure 4. Nine-notebook progression from platform validation to packaging readiness.

Notebook	Focus	Primary work	Gate/output
01	Platform readiness	Runtime, GPU, storage, dependencies and paths	Environment readiness
02	Data engineering	Integrity, EDA, transformations and class weights	Data readiness
03	Model development	ResNet-18 training, checkpointing and thresholds	Model bundle
04	Evaluation	Isolated test inference, metrics and errors	Frozen test evidence
05	Explainability	LayerGradCam and representative cases	Visual evidence registry
06	Grounded language	FLAN-T5 fine-tuning, evaluation and guardrail	Language bundle
07	FastAPI	Schemas, services, routes and tests	12-path API
08	Streamlit	HTTP client, state, rendering and live validation	Demonstration UI
09	Packaging	Containers, Compose, documentation and audit	Final readiness

5. Platform Readiness and Reproducibility

Notebook 1 validates a Kubeflow/Jupyter runtime with Python 3.11, an NVIDIA A100-SXM4-80GB GPU, CUDA 12.1 and PyTorch 2.5.1+cu121. It separates lightweight solution source code from datasets, checkpoints, model bundles, Hugging Face caches, MLflow runs and generated outputs stored on the dedicated data volume.

Control	Implementation	Evidence
GPU	NVIDIA A100-SXM4-80GB	Available and CUDA-enabled
Precision	FP16/BF16 functional checks	Passed
Path registry	configs/paths.yaml	Created and reused
Experiment tracking	File-based MLflow URI	Routed to persistent volume
Storage reserve	4.00 GiB protected	8.71 GiB free at final audit
Final gate	Environment readiness	Ready for data engineering

Centralized configuration prevents later notebooks from silently diverging on label ordering, image resolution, storage locations or model versions. Cache routing also avoids exhausting the root filesystem during model retrieval and full-model fine-tuning.

6. Dataset Engineering and Exploratory Analysis

ChestMNIST is a standardized multilabel chest X-ray dataset with fourteen binary findings. The project uses the 224 x 224 archive and preserves the official train, validation and test partitions. Every image can have zero, one or multiple positive labels; therefore the target is a fourteen-element binary vector rather than a single softmax class.

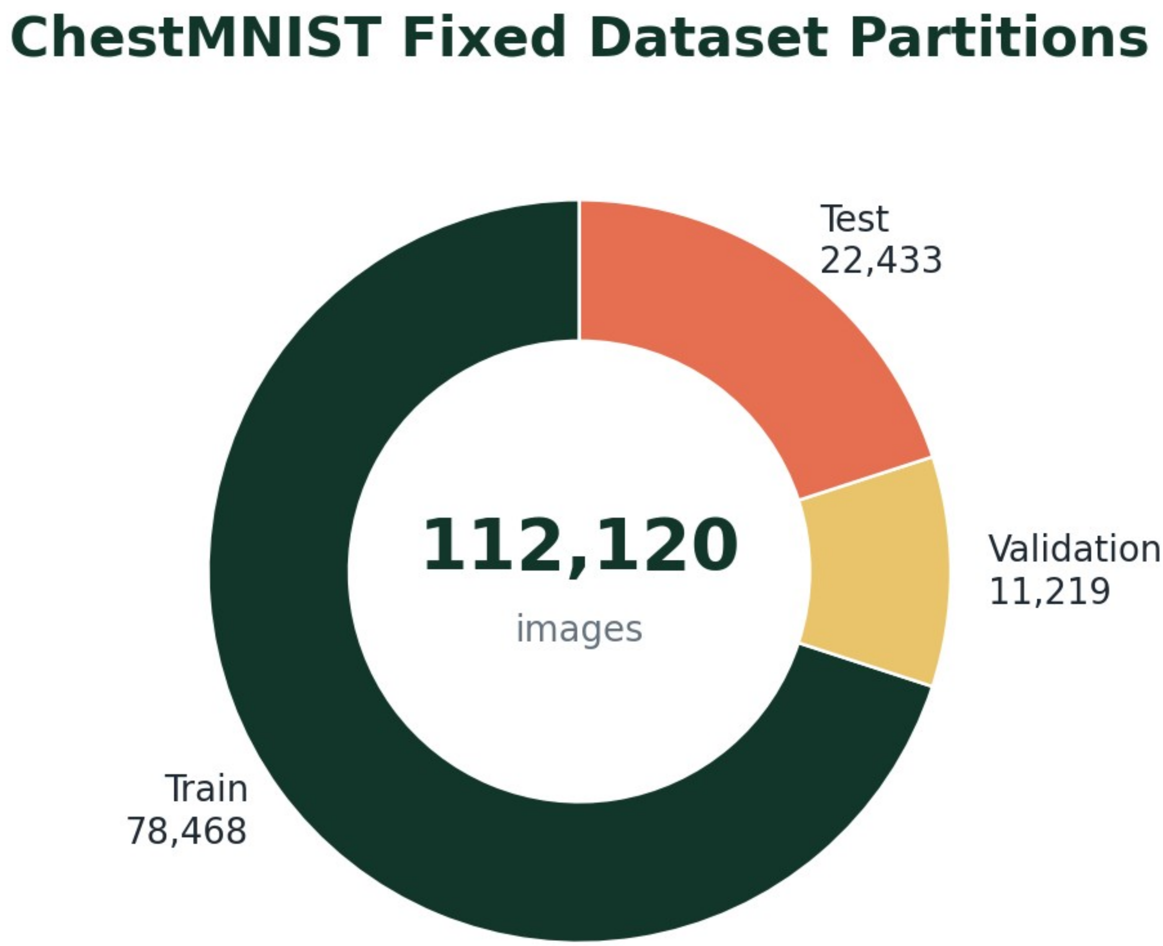

Figure 5. Fixed ChestMNIST train, validation and isolated-test partitions. Source: Notebook 2.

Label ID	Finding
0	Atelectasis
1	Cardiomegaly
2	Effusion
3	Infiltration
4	Mass
5	Nodule
6	Pneumonia
7	Pneumothorax
8	Consolidation
9	Edema

Label ID	Finding
10	Emphysema
11	Fibrosis
12	Pleural abnormality
13	Hernia

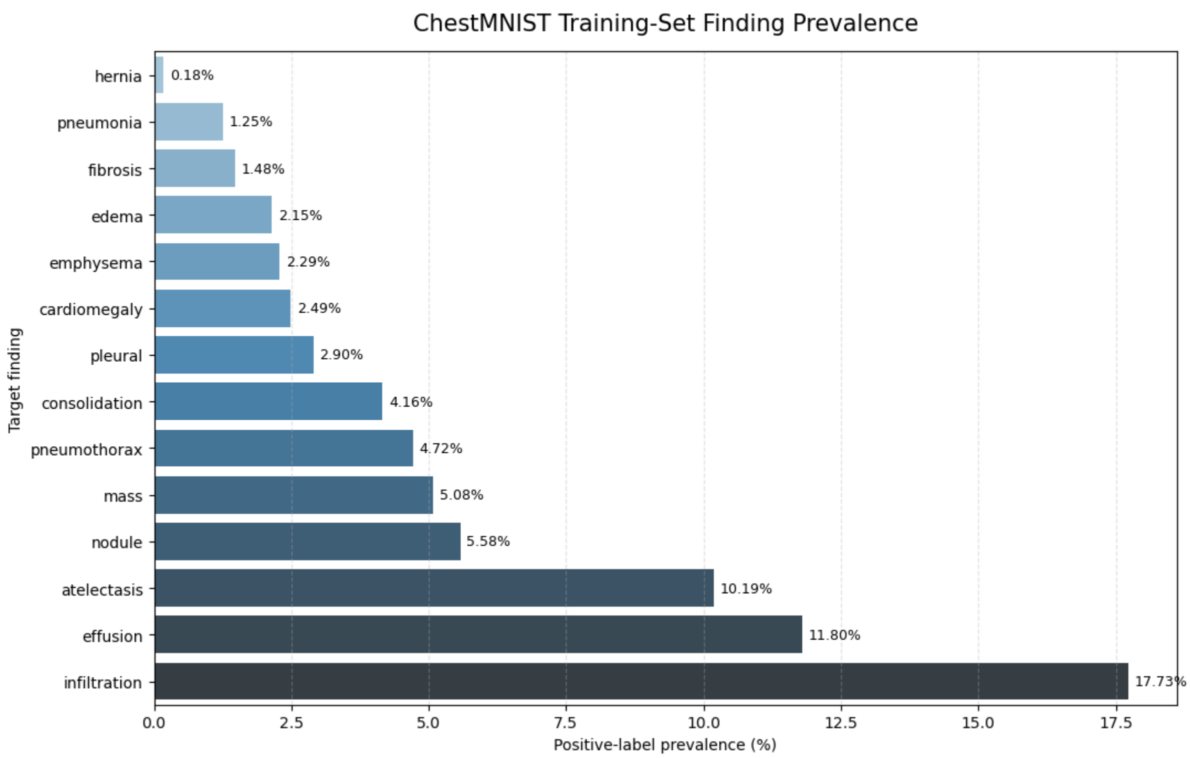

Figure 6. Training-set prevalence across the fourteen supported findings. Source: Notebook 2 embedded output.

The prevalence plot demonstrates strong class imbalance. Infiltration is the most frequent training finding at 17.73%, followed by effusion at 11.80% and atelectasis at 10.19%. Hernia is extremely rare at 0.18%. Positive-class weights therefore range from 4.64 for infiltration to a raw 543.92 for hernia; the hernia weight is capped at 100 to control optimization instability.

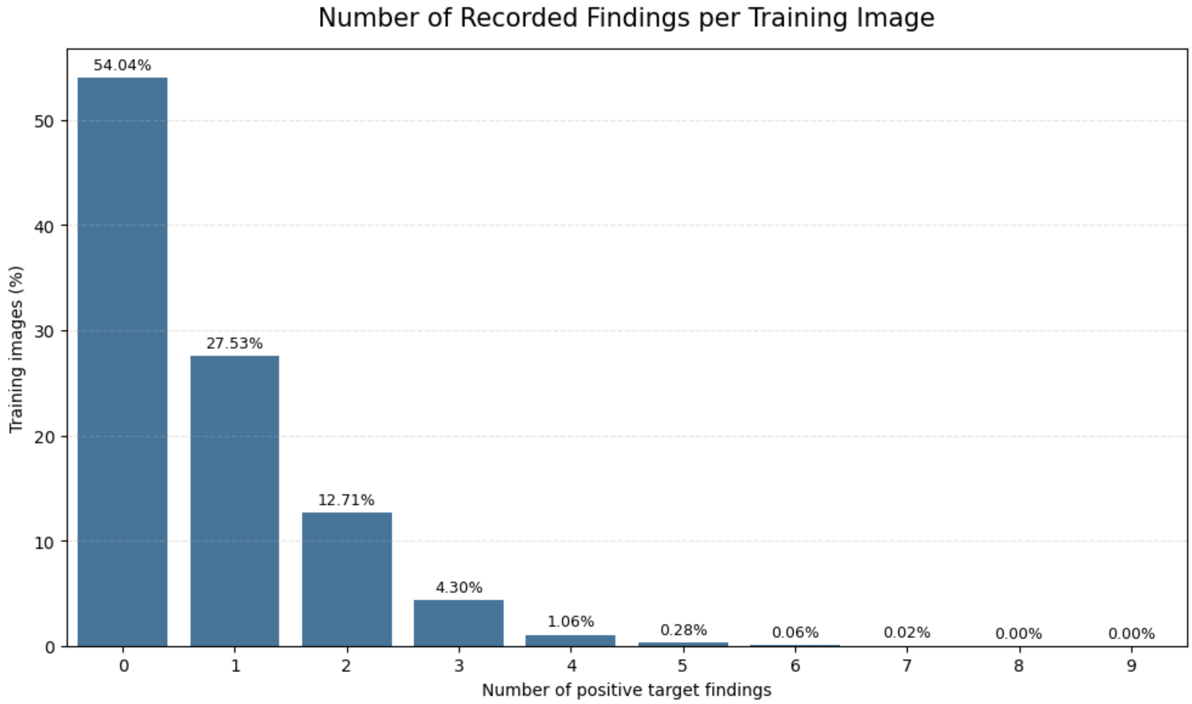

Figure 7. Distribution of no-target, single-finding and multiple-finding records. Source: Notebook 2 embedded output.

Within the training split, 54.04% of images have no target finding, 27.53% have exactly one finding and 18.43% contain multiple findings. The mean label cardinality is 0.720, confirming that both sparse positive labels and co-occurring findings must be handled explicitly.

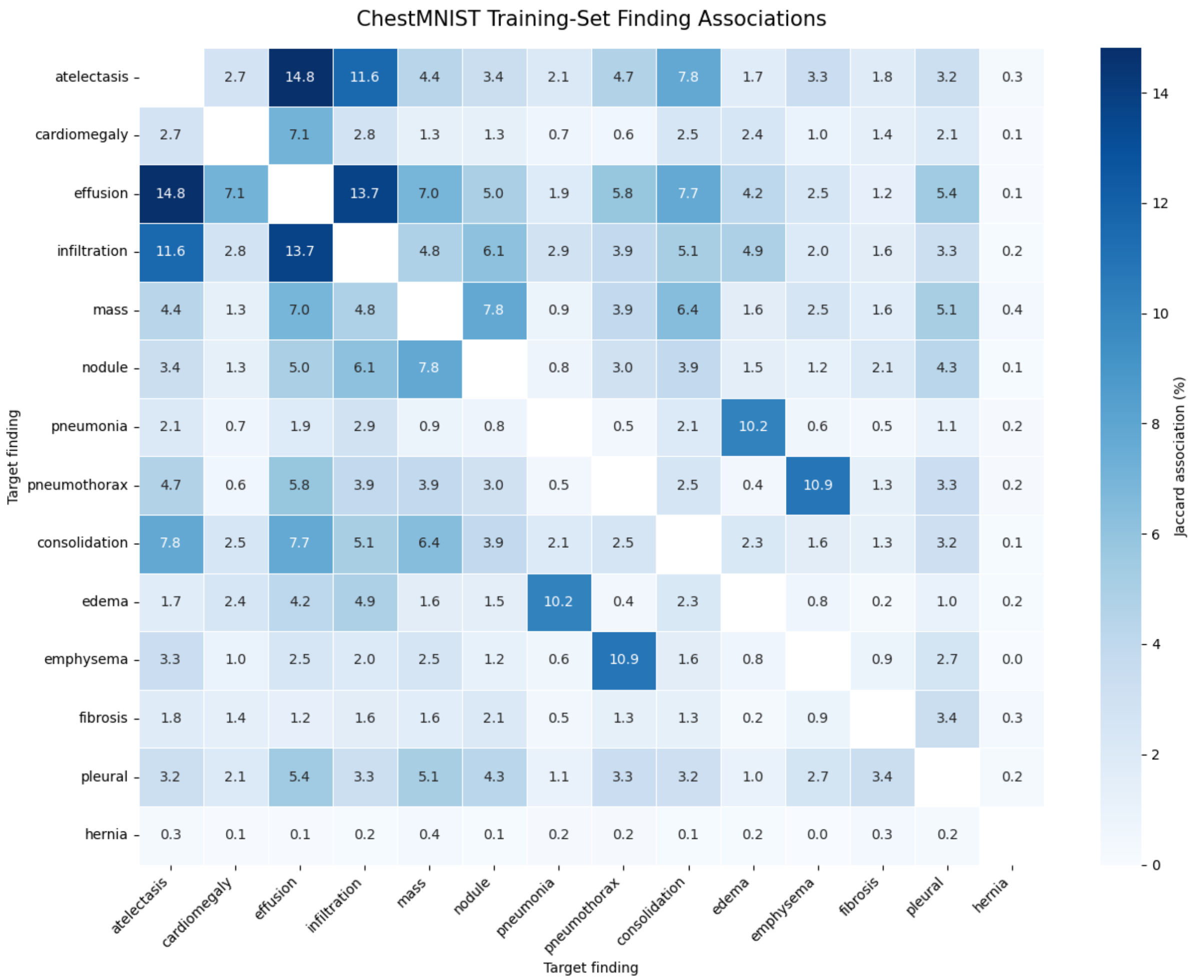

Figure 8. Finding co-occurrence counts and Jaccard association heatmap. Source: Notebook 2 embedded output.

The strongest observed pairs include atelectasis-effusion, effusion-infiltration and atelectasis-infiltration. Association analysis supports multilabel modeling because the findings are not mutually exclusive. It also motivates representative-case evaluation instead of relying only on one aggregate score.

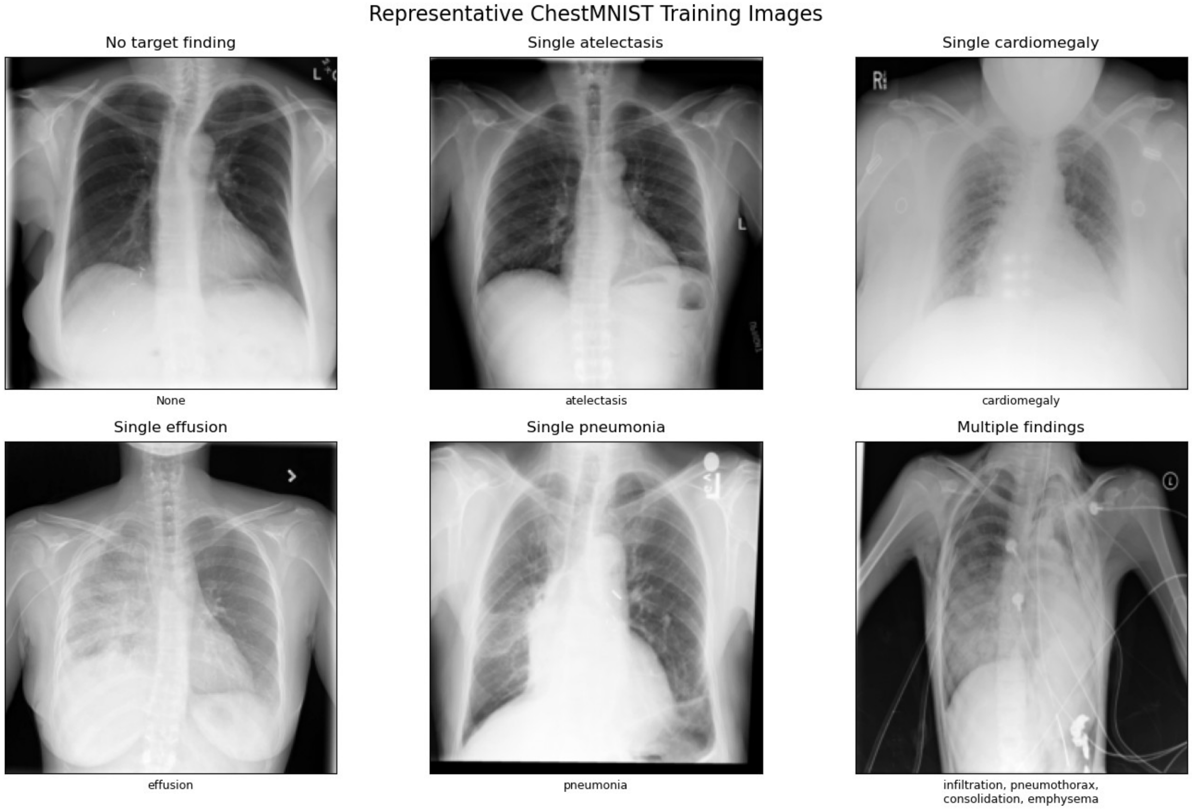

Figure 9. Representative ChestMNIST images covering no-target, single-finding and multiple-finding cases. Source: Notebook 2 embedded output.

Training augmentation is deliberately conservative for radiographs: random affine transformations are limited to +/-5 degrees, 2% translation and 0.95-1.05 scaling, while brightness and contrast jitter are limited to 0.9-1.1. All inputs are converted to three channels and normalized with ImageNet statistics to match the pretrained ResNet-18 contract.

7. Computer-Vision Model Development

The classifier starts from ImageNet-pretrained ResNet-18 and replaces the final classification layer with fourteen independent logits. Sigmoid converts each logit to a probability. BCEWithLogitsLoss receives the positive-class weights produced by Notebook 2 so rare findings contribute meaningfully during training.

Model-development element	Persisted value
Backbone	ResNet-18 pretrained on ImageNet
Output	14 independent logits / sigmoid probabilities
Training records	78,468
Validation records	11,219
Precision	BF16 mixed precision on A100

Model-development element	Persisted value
Epochs	5 completed; epoch 2 selected
Best validation macro ROC-AUC	0.8210
Elapsed training duration	7.34 minutes
Tracking	MLflow experiment chestmnist-resnet18-development

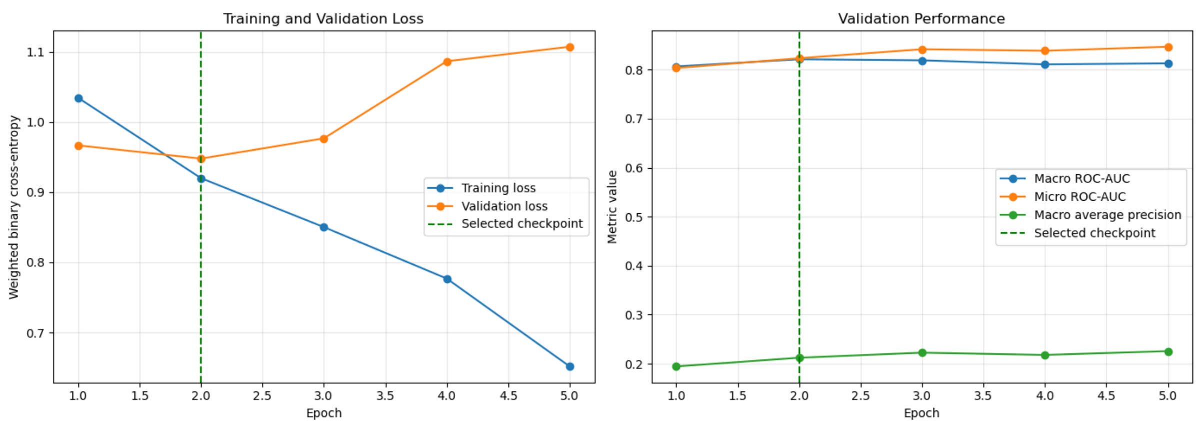

Figure 10. ResNet-18 training loss and validation macro ROC-AUC across five epochs. Source: Notebook 3 embedded output.

Training loss decreases consistently, while validation loss reaches its minimum and macro ROC-AUC reaches its maximum at epoch 2. Subsequent training improves the fit to training data but reduces the selected validation objective, so the versioned checkpoint remains epoch 2. This is evidence of validation-based model selection rather than selection on the isolated test set.

Finding-specific thresholds are calibrated on validation data before test evaluation. Examples include 0.6758 for effusion, 0.8281 for pneumothorax, 0.8750 for emphysema and 0.6992 for hernia. Threshold calibration raises validation macro F1 from 0.2024 at a uniform 0.50 decision boundary to 0.2947, while Hamming loss improves from 0.2841 to 0.0949.

Frozen inference contract: Model weights, label order, preprocessing, thresholds and version metadata are exported together as resnet18-chestmnist-v1 before isolated-test evaluation.

8. Isolated Test Evaluation and Error Analysis

Notebook 4 restores the versioned model bundle and applies it to 22,433 isolated test images without modifying weights or thresholds. The evaluation deliberately reports both ranking metrics and threshold-dependent decision metrics because class imbalance can make any single number incomplete.

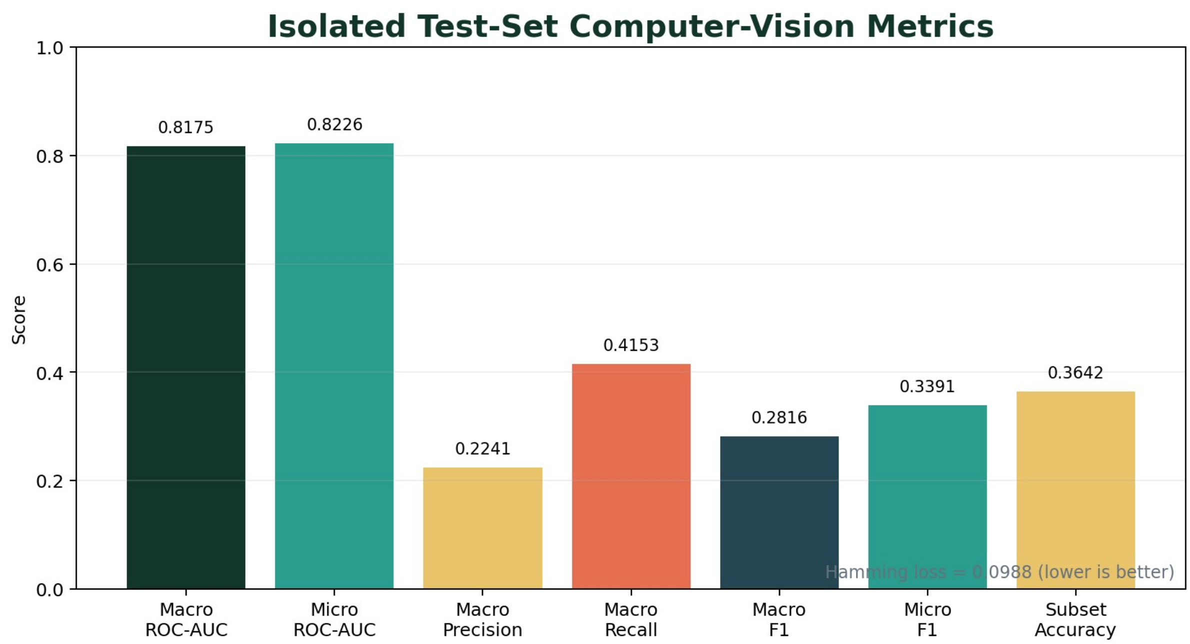

Figure 11. Overall isolated test-set performance. Source: Notebook 4.

Metric	Value	Interpretation
Macro ROC-AUC	0.8175	Equal weight to each finding's ranking quality
Micro ROC-AUC	0.8226	Aggregate ranking across all label decisions
Macro average precision	0.2216	Imbalance-sensitive precision-recall summary
Macro precision	0.2241	Average positive predictive precision
Macro recall	0.4153	Average sensitivity across findings
Macro F1	0.2816	Harmonic balance of macro precision and recall
Micro F1	0.3391	Global label-level precision/recall balance
Hamming loss	0.0988	Fraction of incorrect individual label decisions
Subset accuracy	0.3642	Exact match across all fourteen labels

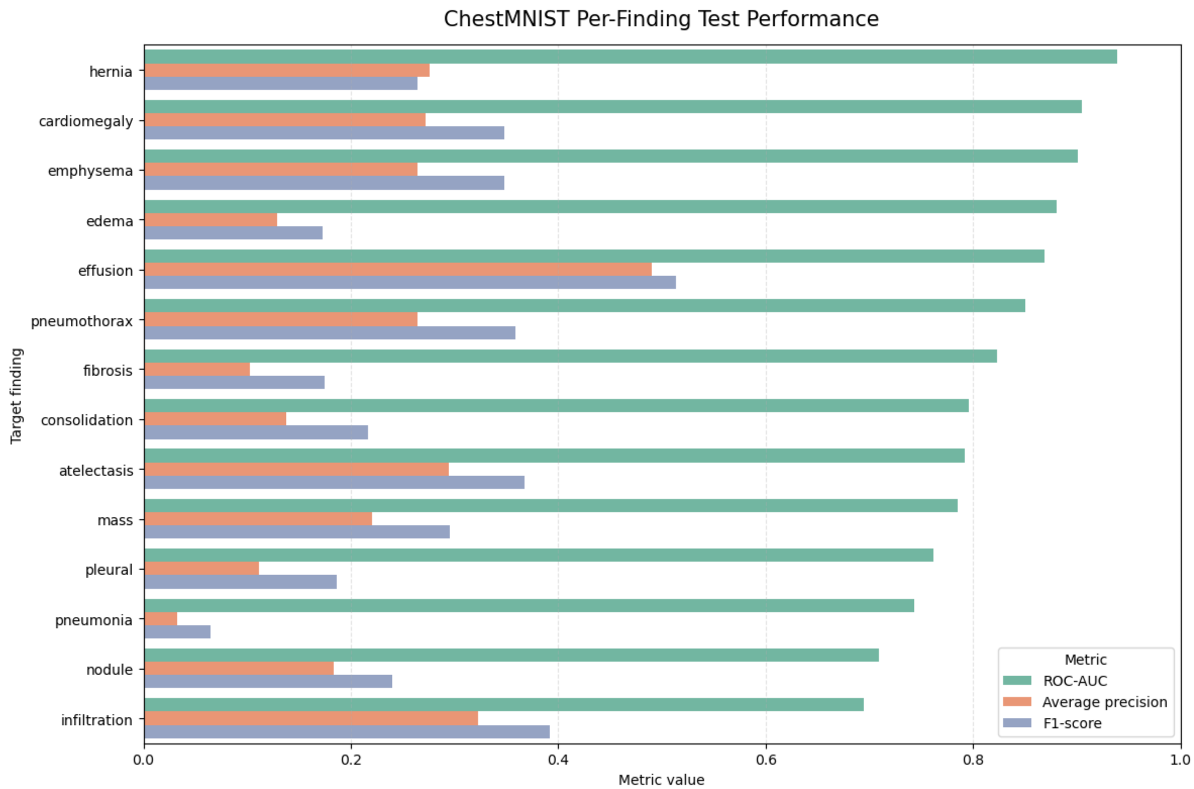

Figure 12. Per-finding ROC-AUC, average precision and F1 comparison. Source: Notebook 4 embedded output.

Per-finding discrimination is strongest for hernia (ROC-AUC 0.9385), cardiomegaly (0.9046), emphysema (0.9011), edema (0.8799) and effusion (0.8687). Infiltration has the lowest ROC-AUC at 0.6944. Average precision is substantially lower for rare labels, demonstrating why prevalence-aware interpretation is required.

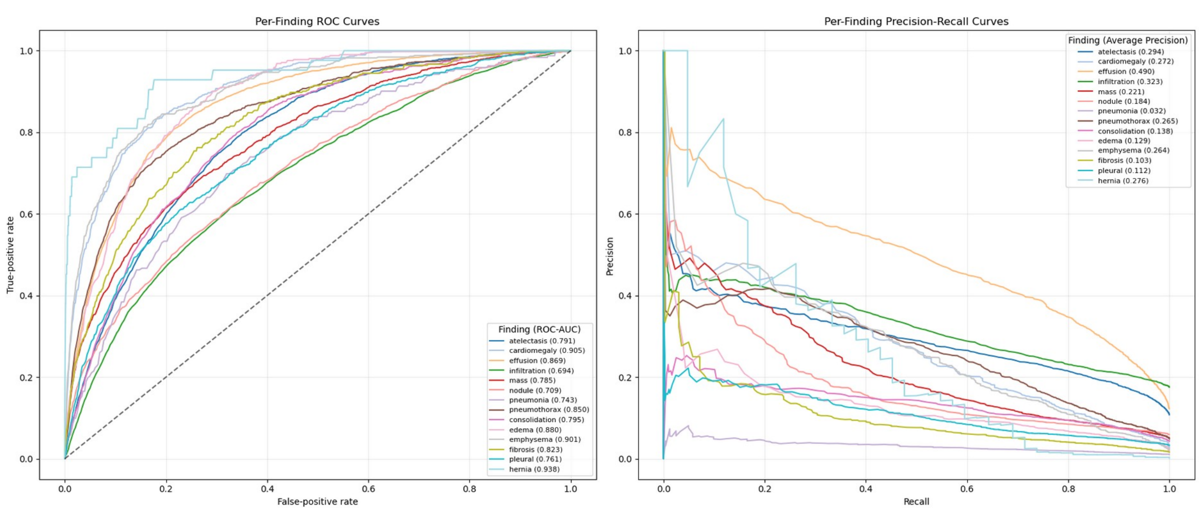

Figure 13. ROC and precision-recall curves for all fourteen findings. Source: Notebook 4 embedded output.

ROC curves show ranking performance across thresholds, whereas precision-recall curves expose the practical difficulty of rare positive findings. The divergence between ROC-AUC and average precision for several findings is expected under severe class imbalance and prevents overstating performance.

Figure 14. Finding-specific confusion matrices under frozen calibrated thresholds. Source: Notebook 4 embedded output.

The confusion matrices make operating-point trade-offs explicit. For example, effusion achieves 1,920 true positives and 2,806 false positives, while pneumonia achieves only 55 true positives against 1,407 false positives because it is rare and difficult. These results are reported transparently rather than hidden behind aggregate metrics.

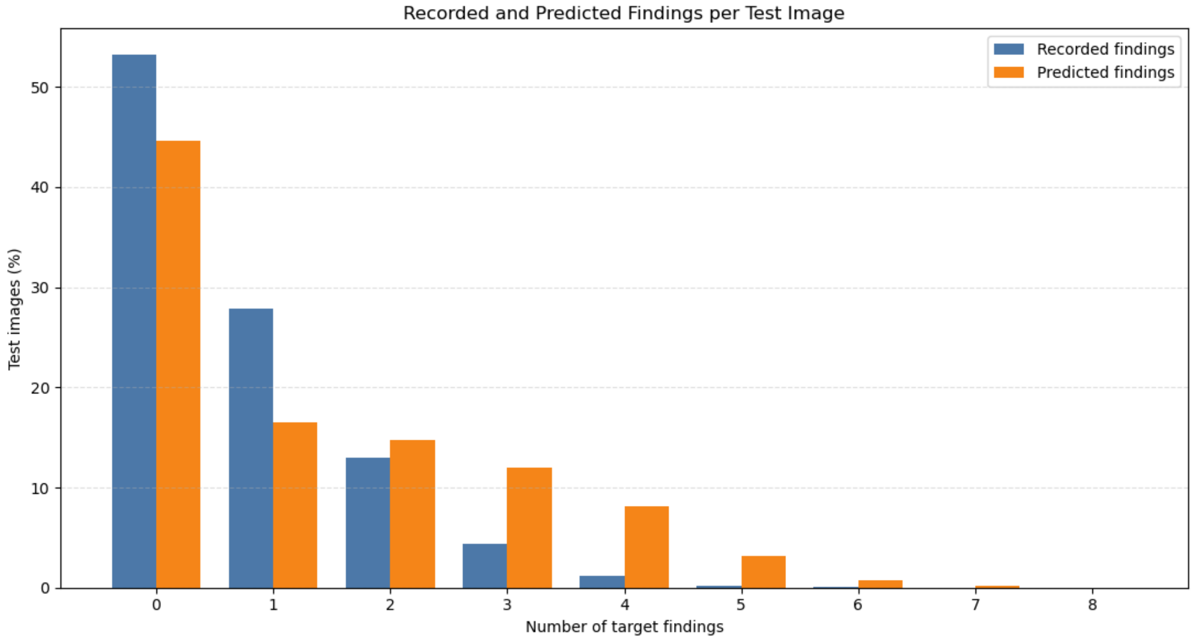

Figure 15. Recorded versus predicted finding-cardinality distribution. Source: Notebook 4 embedded output.

The classifier predicts no finding for 44.59% of test images versus 53.17% in the reference labels, and predicts two or more findings more frequently than recorded. Exact fourteen-label agreement is 36.42%; 37.27% of reference no-finding images receive at least one model alert. This error profile reinforces the educational-use limitation and the need for qualified review.

9. Visual Explainability with Grad-CAM

Notebook 5 uses Captum LayerGradCam on the final convolutional layer, layer4[-1].conv2. The attribution target is finding-specific: gradients are computed for one selected output logit rather than for a generic image-level decision. Positive attributions are normalized and rendered as heatmaps and overlays.

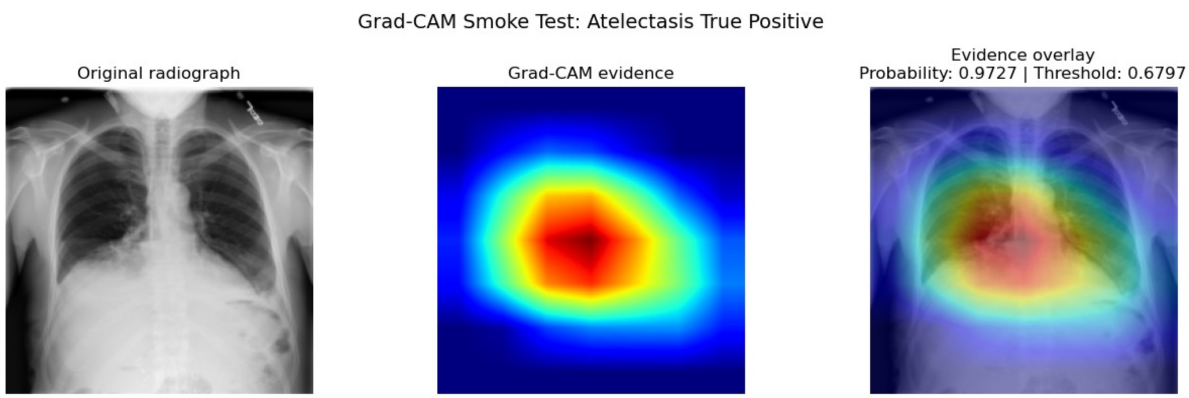

Figure 16. Attribution integrity smoke test showing source image, heatmap and overlay. Source: Notebook 5 embedded output.

Integrity checks verify output dimensions, finite attribution values, normalization bounds and preservation of the frozen inference contract. The visual is interpreted only as evidence of regions that influenced the model output.

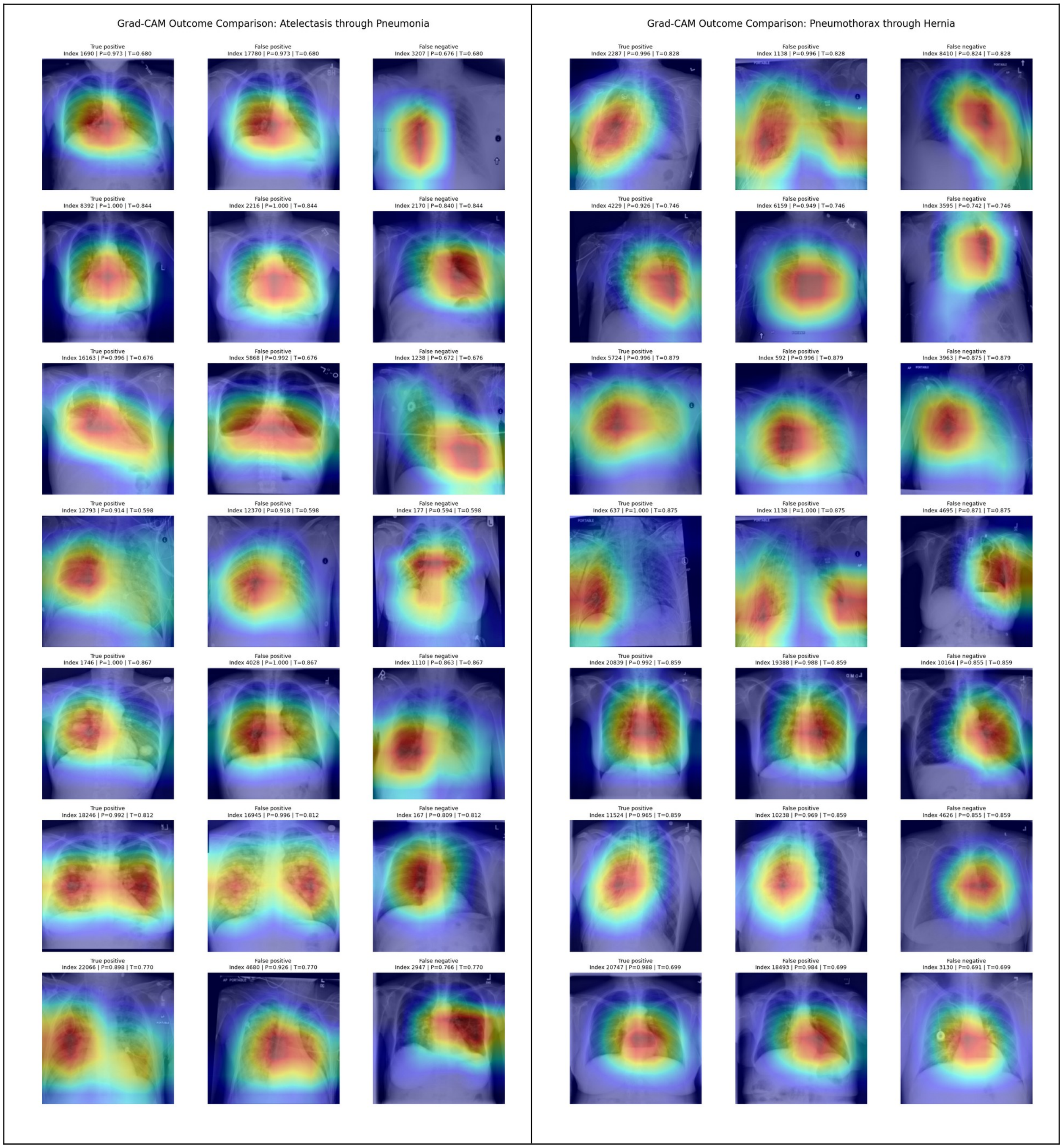

Figure 17. Complete cross-outcome Grad-CAM galleries covering representative true-positive, false-positive and false-negative evidence for all fourteen findings. Source: Notebook 5 persisted outputs.

Interpretation boundary: Grad-CAM does not segment a lesion, confirm an anatomical abnormality, establish causality or provide a clinical diagnosis. It explains model influence, not medical truth.

10. Grounded Language Model and Guardrails

Notebook 6 defines a versioned ontology, prompt registry and four language tasks. A leakage-resistant dataset of 4,800 records is split into 3,600 training, 600 validation and 600 held-out test records. FLAN-T5-small is fully fine-tuned for five epochs; the best validation loss is 0.1574.

Task	Evaluation records	Contract objective
Structured report	150 held-out	Ordered finding evidence and limitations
Plain-language explanation	150 held-out	Accessible description of thresholds and meaning
Grounded question answering	150 held-out	Question answered only from stored prediction evidence
Educational follow-up	150 held-out	Controlled professional-review guidance

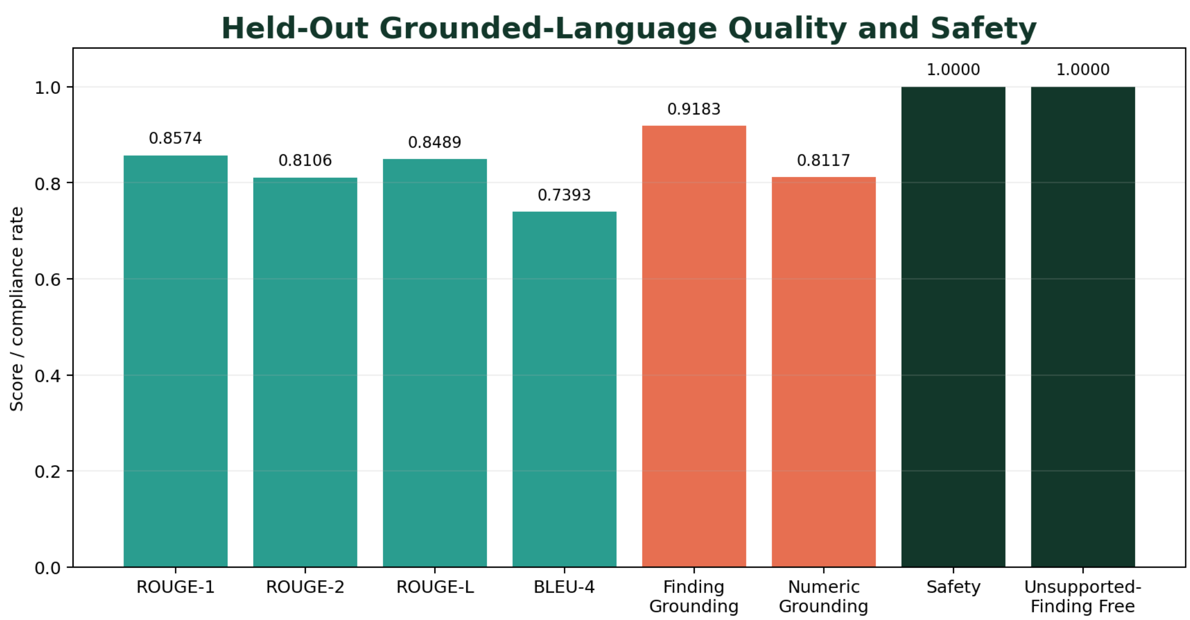

Figure 18. Held-out language quality, grounding and safety metrics. Source: Notebook 6.

Text-overlap quality is high across the controlled reference contract: ROUGE-1 F1 is 0.8574, ROUGE-2 F1 is 0.8106, ROUGE-L F1 is 0.8489 and smoothed BLEU-4 is 0.7393. Exact match is zero because the task permits valid wording variation. More importantly, the unsupported-finding rate is zero, safety-boundary compliance is 1.0000 and forbidden-claim-free rate is 1.0000.

Raw generation still exposes contract weaknesses: finding-grounding compliance is 0.9183 and numeric-grounding compliance is 0.8117. Error analysis identifies 164 of 600 generations with at least one issue, dominated by numeric-grounding issues (113 records) and missing-required-finding issues (49 records).

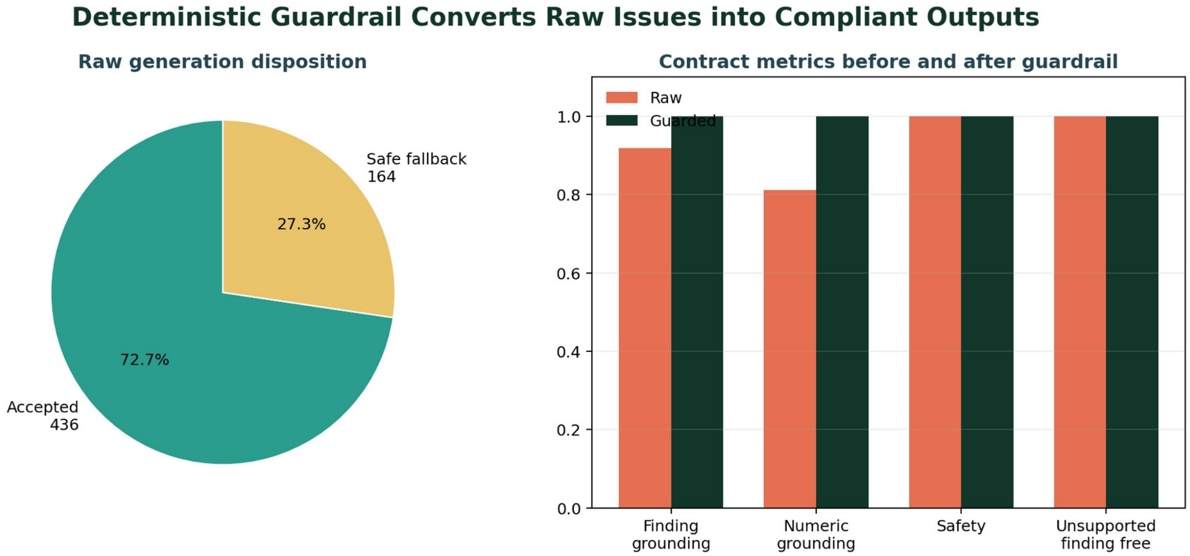

Figure 19. Guardrail disposition and contract compliance before and after deterministic control. Source: Notebook 6.

The deterministic guardrail accepts 436 raw generations and replaces 164 with task-specific safe templates. Post-guardrail evaluation reaches 1.0000 for task routing, section order, required findings, unsupported-finding freedom, numeric grounding, safety limitations, no-target boundaries, controlled refusals, Grad-CAM limitations and forbidden-claim avoidance. The original raw generations are retained for auditability.

11. FastAPI Service and OpenAPI Contract

Notebook 7 converts the model and language artifacts into a reusable service architecture. Pydantic schemas enforce strict request and response contracts; controlled exceptions avoid leaking internal paths; bounded upload reading and format verification protect image endpoints; locks protect inference, Grad-CAM and the in-memory prediction store; middleware records latency and request outcomes.

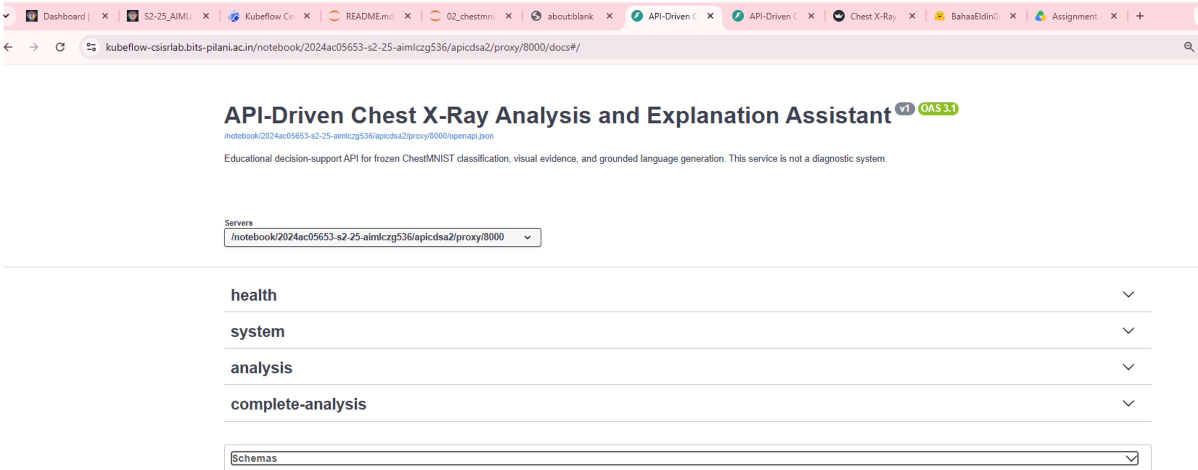

Figure 20. Swagger UI title, OpenAPI 3.1 contract and proxy-aware server configuration. Source: Live FastAPI application.

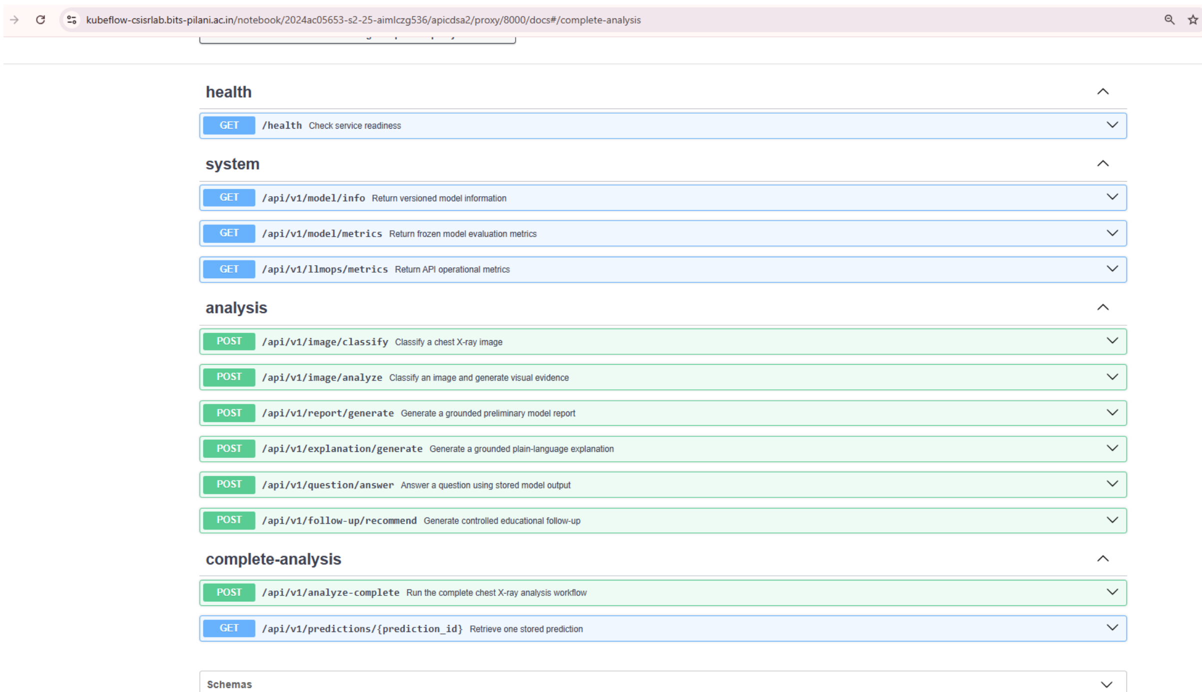

Figure 21. Swagger endpoint catalogue showing all twelve health, system, analysis and complete-analysis operations. Source: Live FastAPI application.

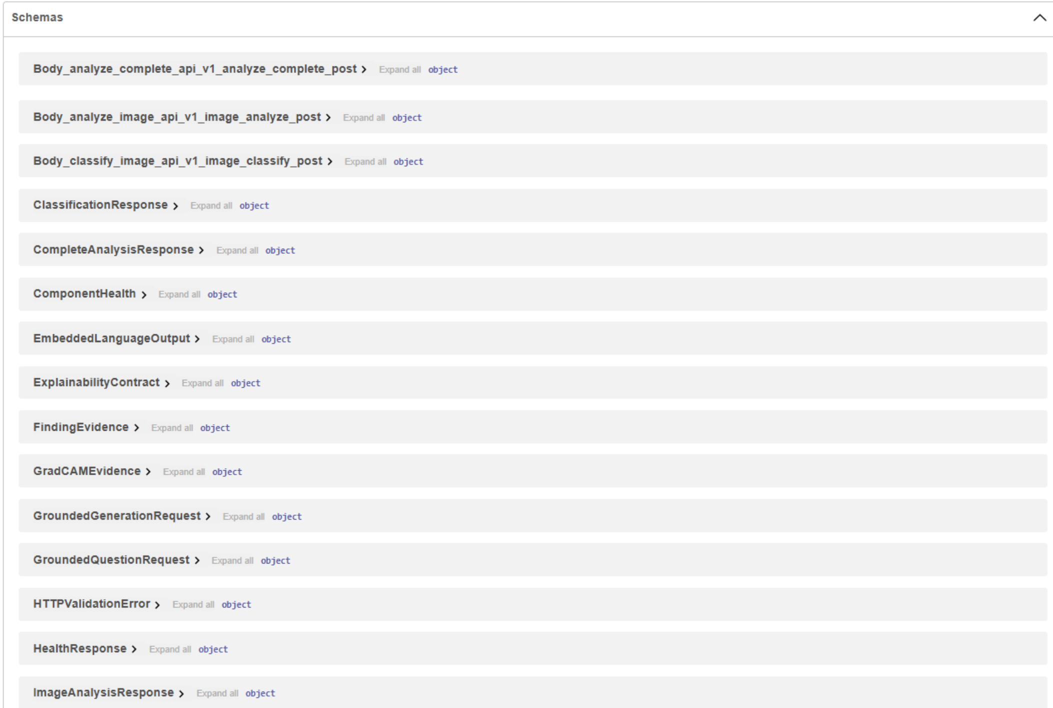

Figure 22. Representative OpenAPI component schemas generated from strict Pydantic contracts. Source: Live FastAPI application.

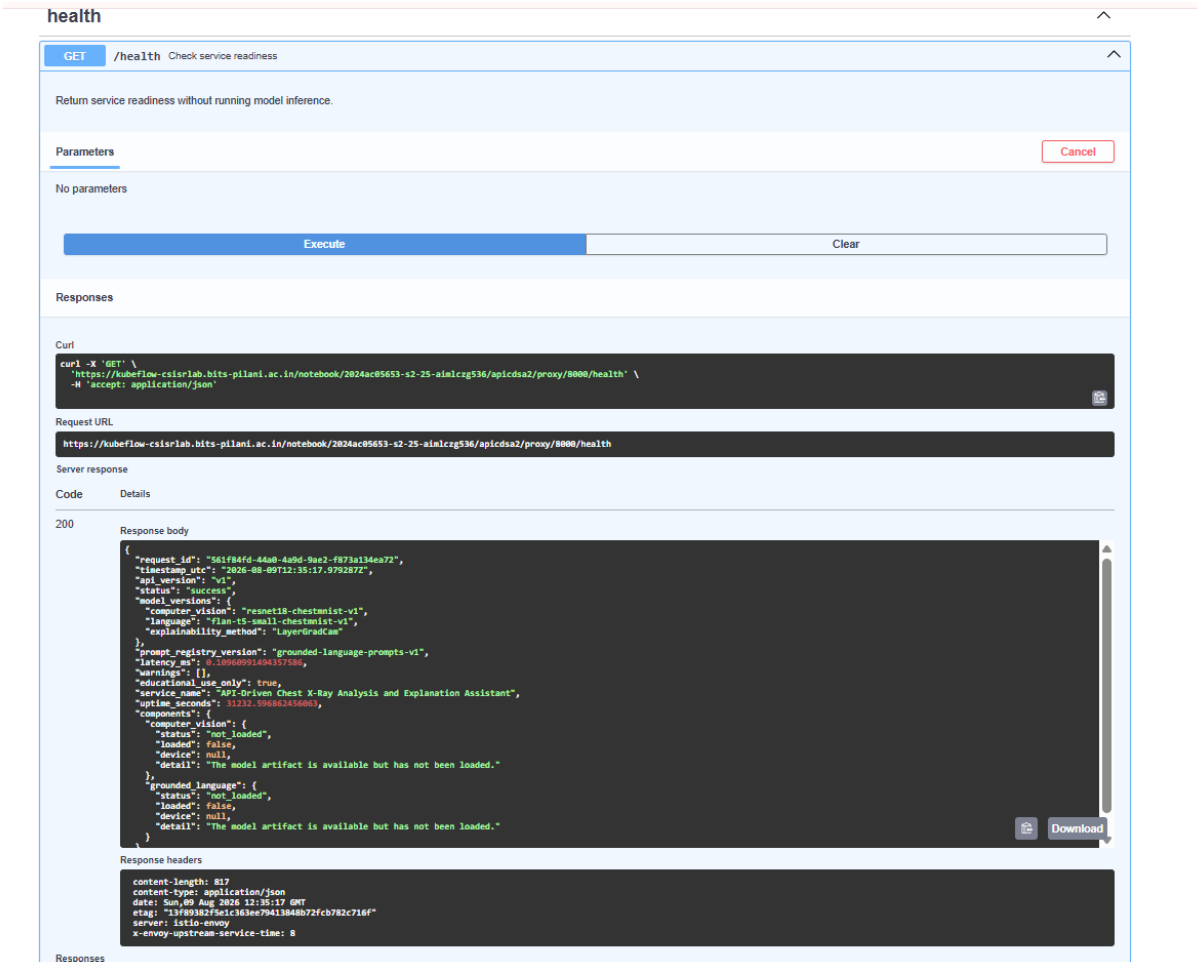

Figure 23. Executed GET /health request with HTTP 200, version lineage, uptime and lazy-load component state. Source: Live FastAPI application.

The health endpoint intentionally returns readiness without loading models. The screenshot therefore shows HTTP 200 and service success while model components remain not_loaded until an inference endpoint requires them. This lazy-load design separates lightweight availability checks from expensive GPU initialization.

API assurance	Evidence
OpenAPI paths	12 / 12
OpenAPI operations	12 / 12
Reusable API tests	23 passed, 0 failed
Standalone Uvicorn	Verified without traceback
Controlled errors	400, 404, 413, 415, 422 and 500 contracts
Artifact registry	45 API files with checksums at Notebook 7 gate

12. Streamlit Interface and Live Demonstration

Notebook 8 builds an HTTP-only Streamlit client. The UI never imports the model, Grad-CAM service or prediction store. It supports PNG/JPEG uploads up to 10 MiB, previews the selected image, sends one complete-analysis request, persists the response in session state, renders structured evidence, submits grounded questions and retrieves stored predictions without repeated inference.

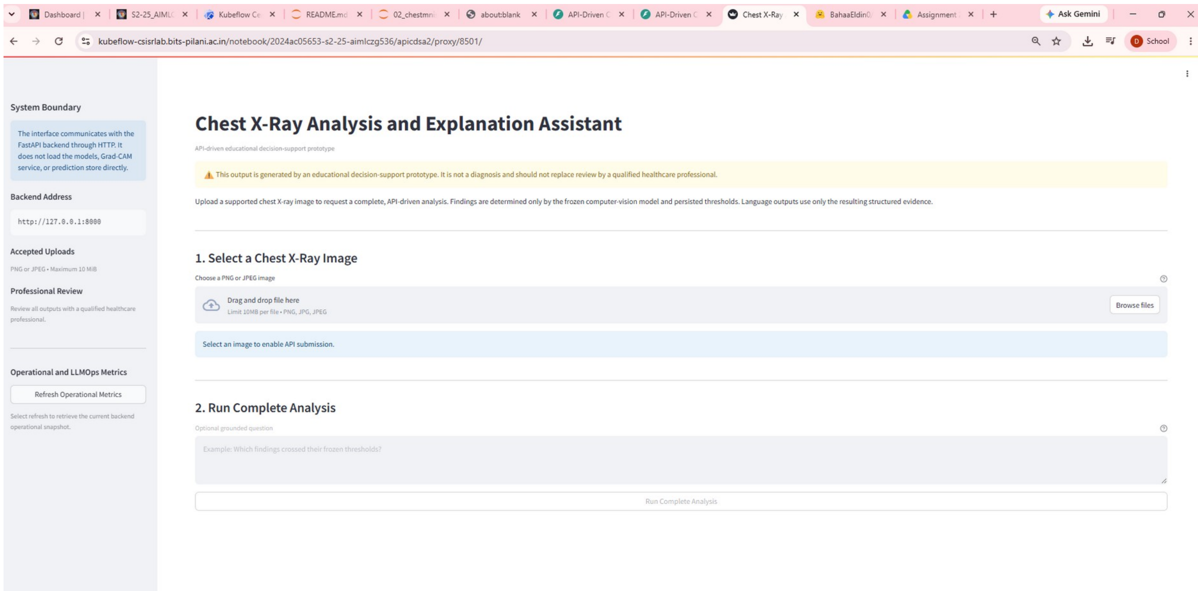

Figure 24. Streamlit landing interface with safety notice, upload contract and HTTP-only system boundary. Source: Live Streamlit application.

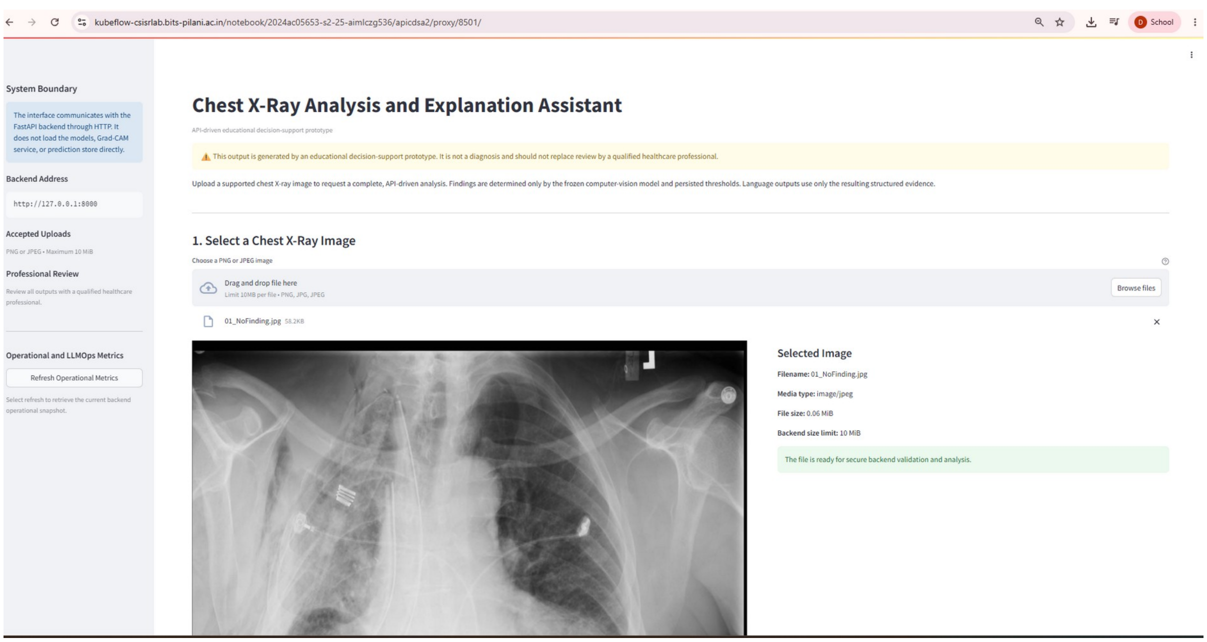

Figure 25. Selected chest X-ray preview and backend-validation boundary before submission. Source: Live Streamlit application.

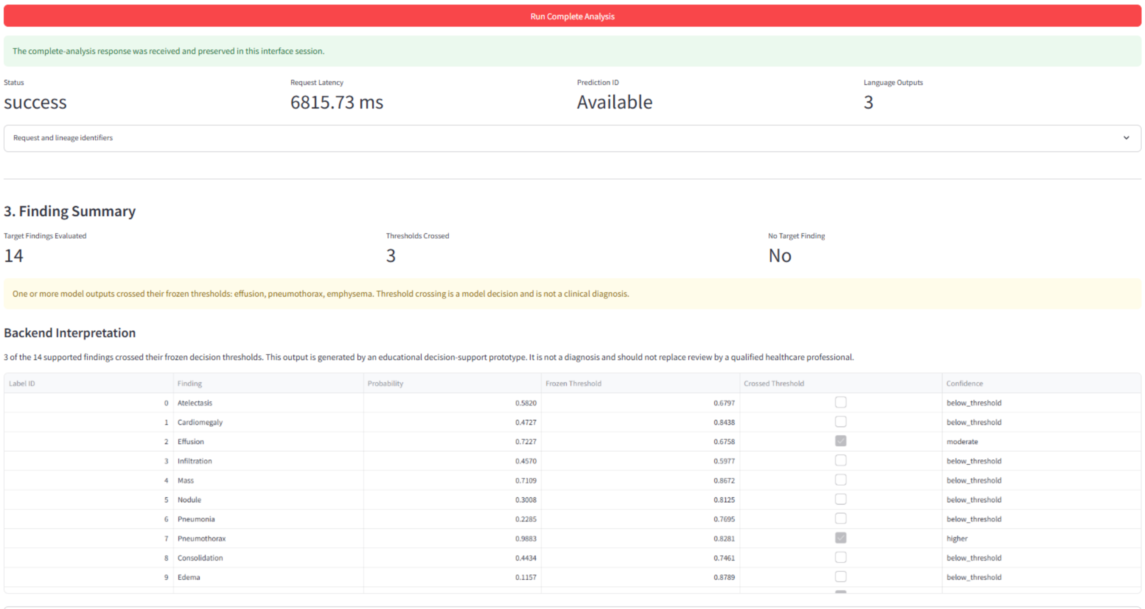

Figure 26. Successful complete-analysis response with 14 findings, three crossed thresholds and persisted prediction ID. Source: Live Streamlit application.

Live demonstration field	Observed application output
Request status	success
End-to-end latency	6,815.73 ms
Findings evaluated	14
Thresholds crossed	3
Crossed model outputs	Effusion, pneumothorax, emphysema
Language outputs	3 embedded tasks

Scope of interpretation: The report documents observed model outputs and workflow behavior. It does not claim that the crossed thresholds are clinically correct for the uploaded image.

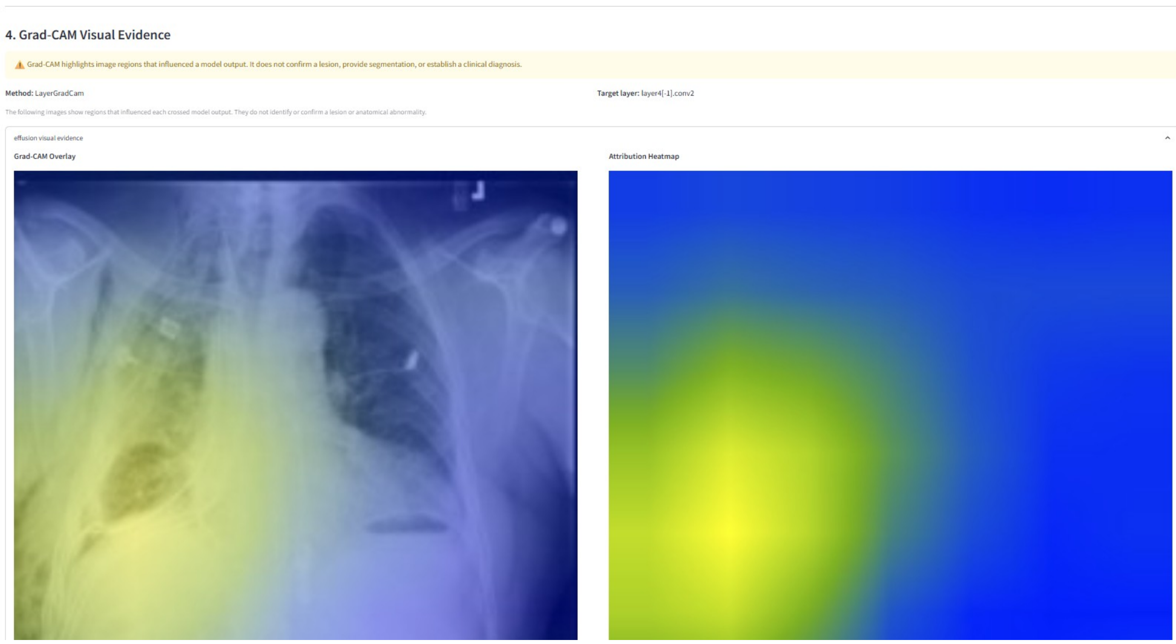

Figure 27. Live effusion Grad-CAM overlay and attribution heatmap with adjacent safety limitation. Source: Live Streamlit application.

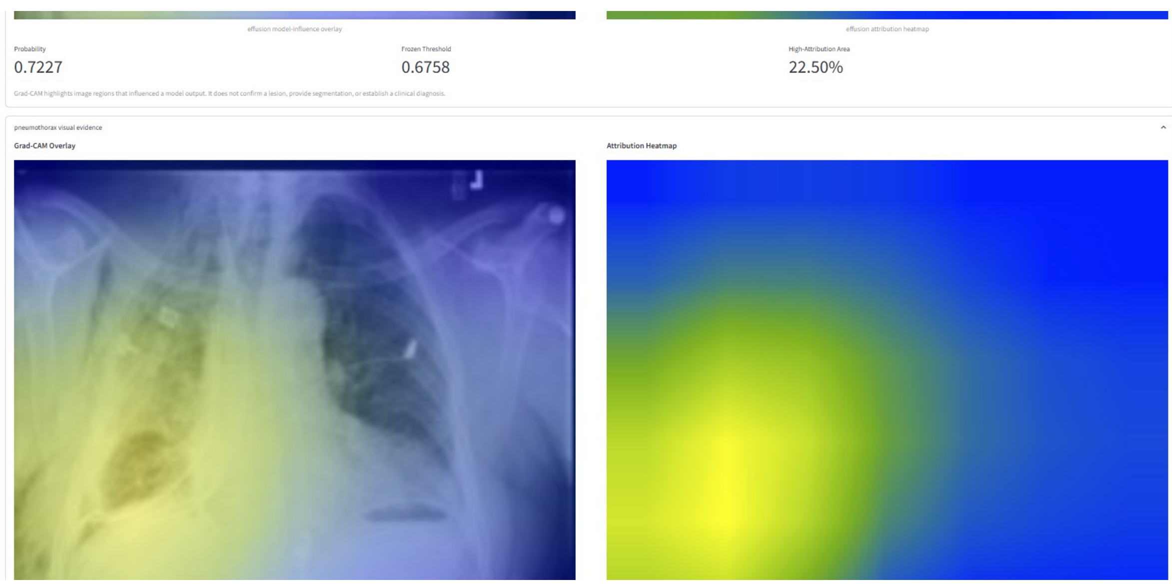

Figure 28. Live visual-evidence panels and evidence metadata for crossed model outputs. Source: Live Streamlit application.

The live application displays Grad-CAM only for crossed findings. Probability, frozen threshold and high-attribution-area percentage remain adjacent to the image evidence so the user can distinguish the decision value from the attribution visualization.

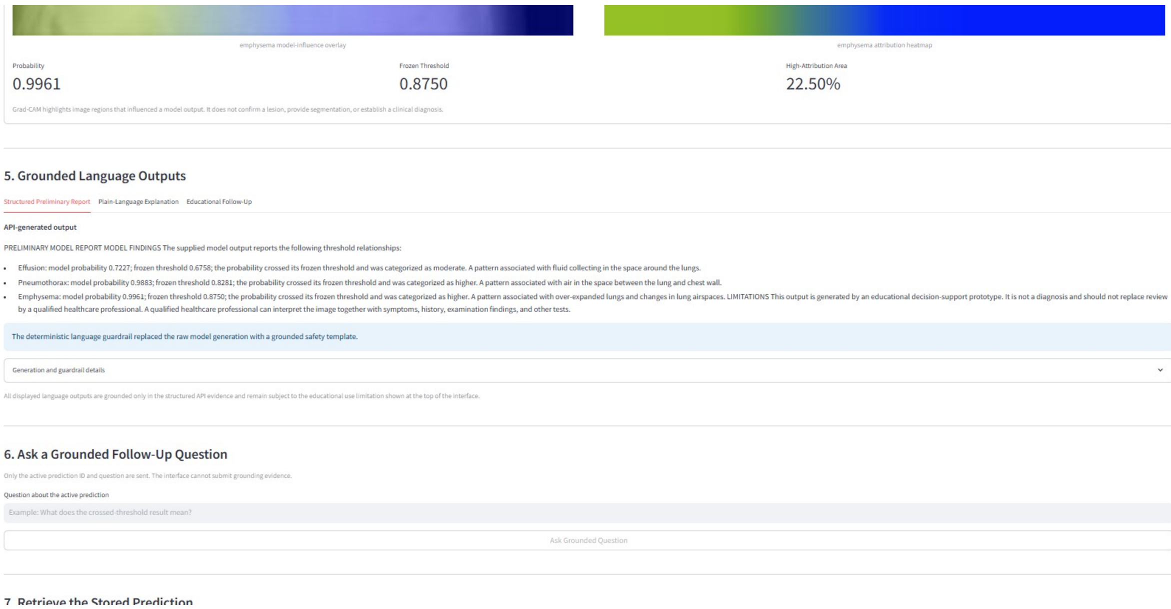

Figure 29. Grounded structured preliminary report and deterministic safety-template notice. Source: Live Streamlit application.

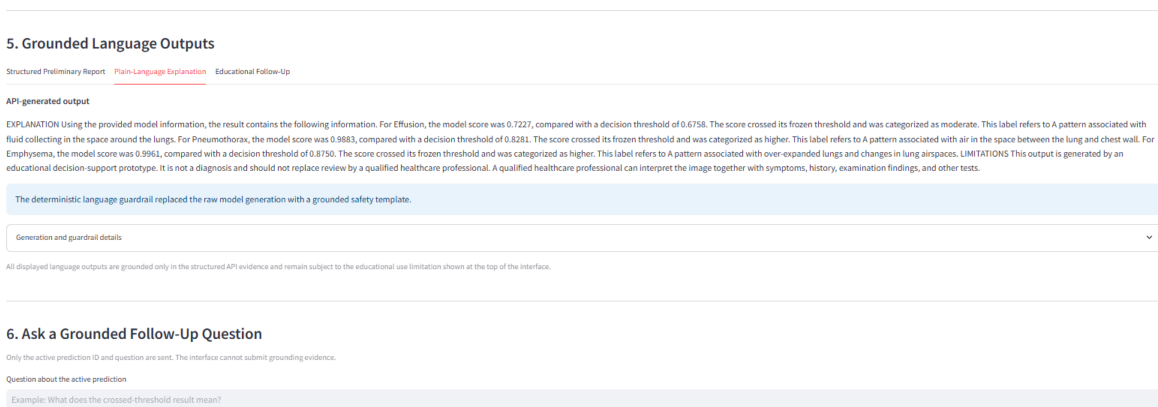

Figure 30. Plain-language explanation grounded in API-provided probabilities and thresholds. Source: Live Streamlit application.

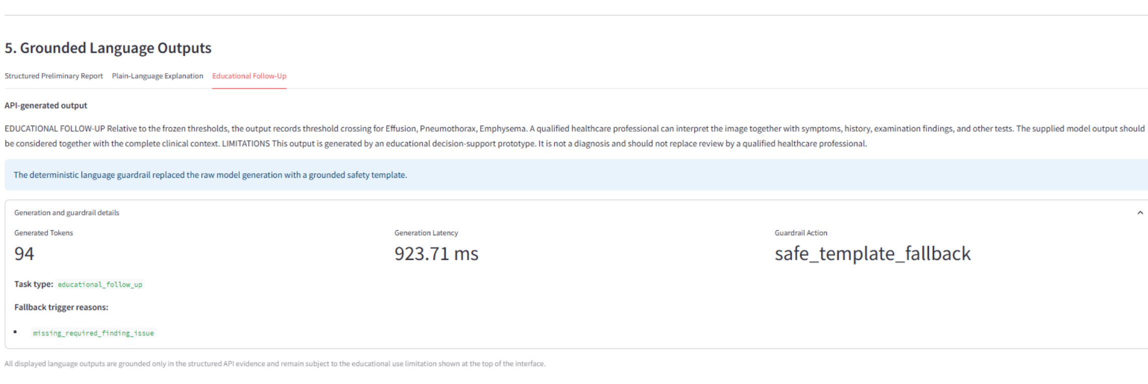

Figure 31. Educational follow-up output with guardrail action, latency and fallback reason. Source: Live Streamlit application.

The main demonstration shows safe_template_fallback for generated language. This is an intended control outcome, not a service failure: the deterministic guardrail detected a missing required finding and replaced the raw output with a grounded template while preserving the trigger reason for audit.

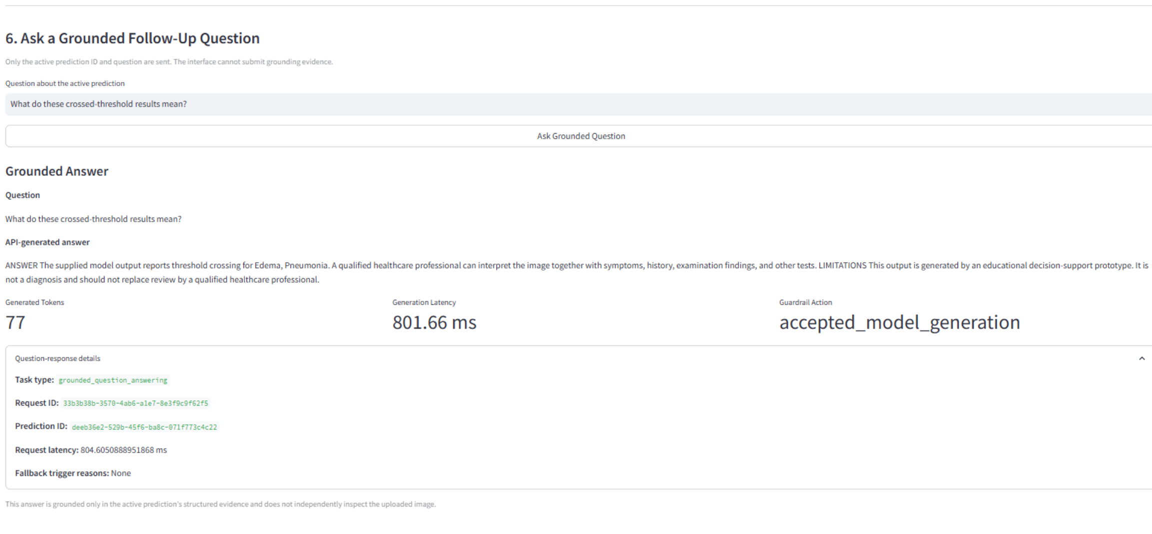

Figure 32. Separate grounded-question interaction showing accepted model generation, request lineage and no fallback. Source: Live Streamlit application.

The follow-up screenshot is treated as a separate prediction interaction because its active evidence differs from the primary demonstration screenshot. It nevertheless validates the same contract: only prediction ID and question are submitted, the answer remains grounded in stored evidence, and the response exposes guardrail action, generated tokens and latency.

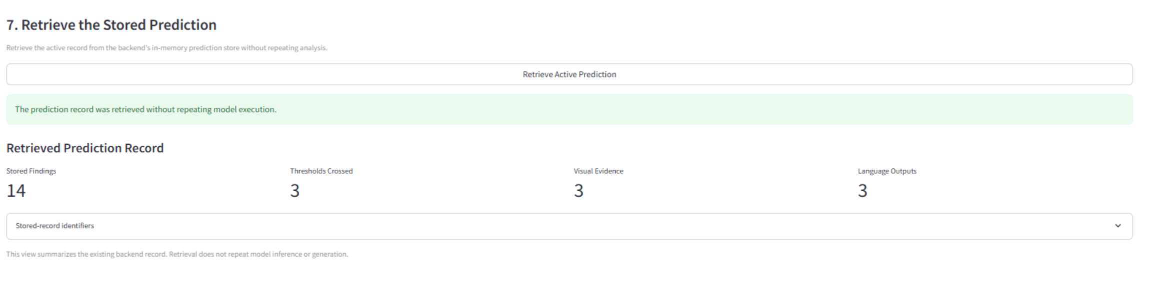

Figure 33. Stored prediction retrieved without repeating classification, Grad-CAM or language generation. Source: Live Streamlit application.

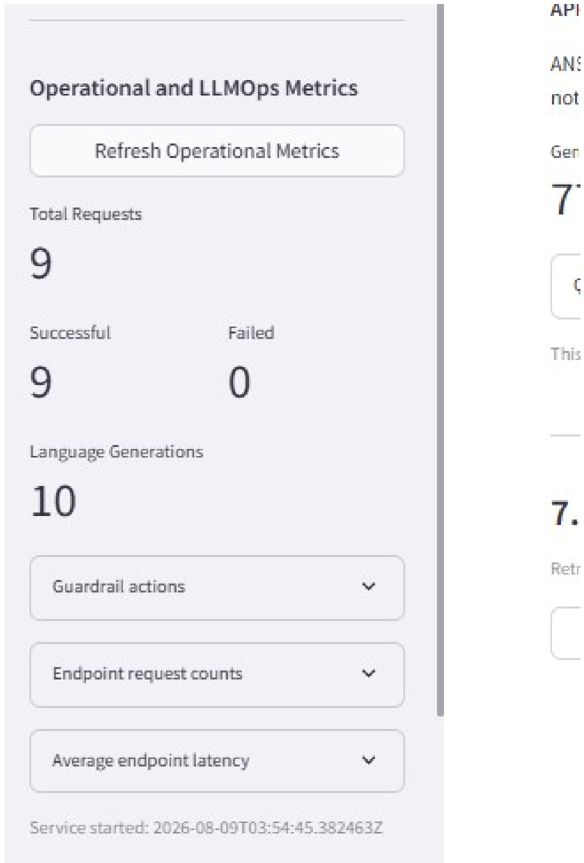

Figure 34. Operational and LLMOps summary showing nine successful requests, zero failures and ten language generations. Source: Live Streamlit application.

Language-generation count can exceed HTTP request count because one complete-analysis request may execute several language tasks. The operational panel therefore measures both request-level and internal language-operation activity.

13. LLMOps, Testing and Operational Evidence

LLMOps is implemented as an evidence chain rather than a single tracking screen. Model, dataset, prompt, schema and API versions are frozen and returned in responses. MLflow records training, evaluation, explainability, language, API, UI and packaging stages. Each stage exports machine-readable evidence and a checksum registry. Operational telemetry records endpoint counts, average latency, language-generation calls and guardrail actions.

LLMOps control	Implementation evidence
Model lineage	resnet18-chestmnist-v1; flan-t5-small-chestmnist-v1
Prompt lineage	grounded-language-prompts-v1
Finding contract	chestmnist-finding-contract-v1
MLflow	Training, evaluation, explainability, language, API, UI and packaging runs

LLMOps control	Implementation evidence
Artifact integrity	SHA-256 checksum registries; 49 final registered files
Operational metrics	Counts, latency, failures, language calls and guardrail actions
Safety monitoring	Raw issues, fallback reasons and post-guardrail compliance
Storage governance	4 GiB protected reserve; final reserve available

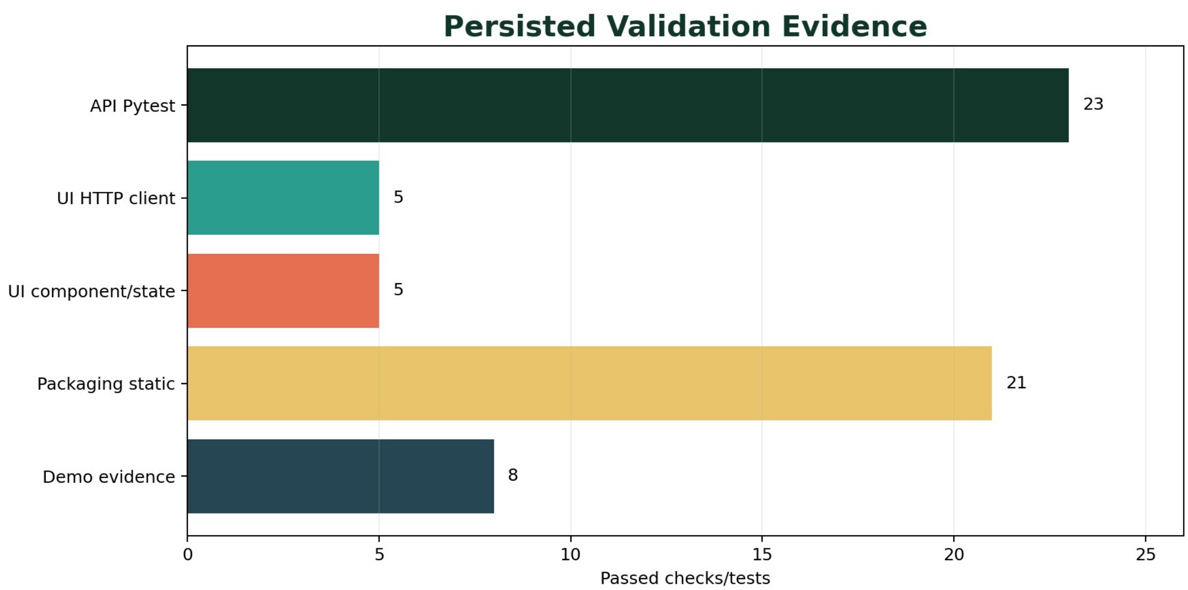

Figure 35. Persisted automated-test and readiness evidence across integration layers.

Validation layer	Passed	Failed	Coverage
API integration	23	0	System, image, language, complete workflow and controlled errors
UI HTTP client	5	0	Health, multipart, controlled errors, validation and timeout
UI component/state	5	0	Initial render, state transitions, table, base64 and HTTP boundary
Static packaging	21	0	Requirements, Dockerfiles, Compose, docs and launch contracts
Demonstration evidence	8	0	API/UI health, screenshots, workflow and artifacts
Final readiness	20	0	End-to-end solution-readiness conditions

14. Packaging and Deployment

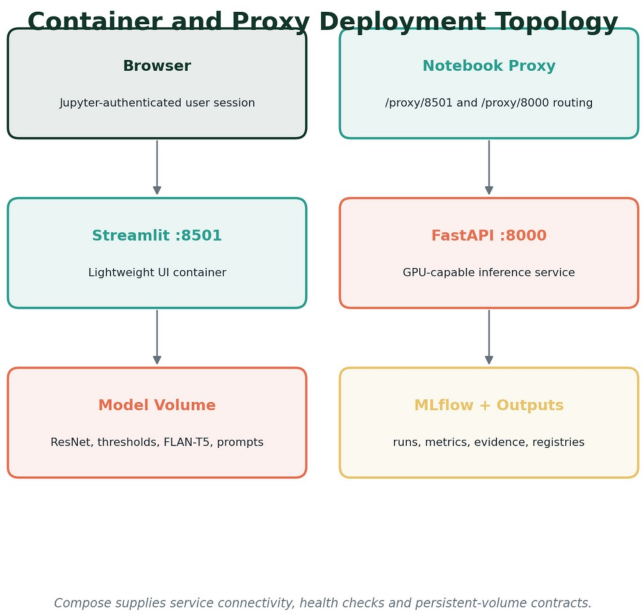

Figure 36. Independent FastAPI and Streamlit deployment topology with proxy routing and persistent volumes.

Notebook 9 defines pinned runtime requirements, separate launch scripts, a GPU-capable FastAPI container, a lightweight Streamlit container, Docker Compose connectivity, persistent model and MLflow volumes, and service-specific health checks. The project intentionally validates static packaging without forcing container builds inside the constrained notebook environment.

Packaging component	Validated contract
FastAPI container	GPU inference dependencies, Uvicorn launch and /health check
Streamlit container	UI-only dependencies and HTTP API base URL
Compose network	Service name resolution and backend-to-UI connectivity
Persistent volumes	Models, outputs, MLflow, caches and artifact registries
Environment contract	API base URL, ports, workers and path configuration
Documentation	README, architecture, usage, API examples, safety and testing
Final registry	49 files with complete checksums
Final MLflow status	FINISHED

Final gate: 20 of 20 final solution-readiness checks passed; static packaging checks 21/21; demonstration checks 8/8; protected storage reserve preserved.

15. Safety, Limitations and Responsible Use

Medical imaging is a high-stakes domain. The project therefore presents the system as an educational decision-support prototype, displays the limitation before any output, propagates educational_use_only through API responses, and requires professional review in all generated language.

Boundary	Responsible interpretation
Dataset labels	ChestMNIST labels are reference labels for benchmark training/evaluation and may contain noise; they are not new clinical diagnoses.
Supported scope	Only fourteen contracted findings are modeled; absence of a crossed threshold does not establish clinical normality.
Threshold decisions	Crossing a frozen threshold is a model operating-point decision, not confirmation of disease.
Grad-CAM	Attribution indicates model influence and is not segmentation, lesion confirmation or causality.
Language	Outputs are grounded in structured evidence but may require safe-template fallback when raw generation violates the contract.
Prediction store	The demonstration store is in-memory and process-local; restarts clear active prediction identifiers.
Generalization	Results are limited to the training domain and were not validated for clinical deployment, demographics, acquisition shifts or external institutions.
Privacy	Demonstration images must remain de-identified; no patient-specific clinical recommendations are produced.

Required wording: Use 'model output', 'dataset reference label', 'crossed threshold' and 'attribution evidence'. Avoid 'diagnosis', 'confirmed lesion', 'normal X-ray' or 'disease detected'.

16. Assignment Compliance Matrix

Assignment requirement	Project evidence	Status
Select a domain	Healthcare / medical imaging	Satisfied
Choose two AI categories	Computer Vision and Natural Language Processing	Satisfied
At least five subtasks	Six integrated subtasks	Exceeded
Identify appropriate models	ResNet-18, LayerGradCam and FLAN-T5-small	Satisfied
Cohesive unified objective	Vision evidence grounds every language output	Satisfied
API-based interactive application	12-path FastAPI service, Swagger UI and Streamlit	Satisfied
Apply LLMOps	MLflow, versioning, registries, monitoring and guardrails	Satisfied
At least five metrics	CV, language, grounding, safety and operational metrics	Exceeded

Assignment requirement	Project evidence	Status
Fine-tune a model	ResNet-18 and FLAN-T5-small fine-tuned	Exceeded
Demonstrate fine-tuned model	Live complete-analysis workflow and screenshots	Satisfied
Screenshots with explanation	Notebook outputs, Swagger and Streamlit figures integrated throughout	Satisfied
Member contributions	Five Assignment II workstreams at 20% each	Satisfied

17. Conclusions and Future Work

The project demonstrates a complete API-driven AI solution rather than an isolated model notebook. It integrates two AI categories and six cohesive subtasks, fine-tunes both vision and language models, exposes strict versioned contracts, presents an interactive interface, preserves evidence and lineage, and validates packaging and operational behavior.

The results also show the value of transparent limitations. Ranking performance is materially stronger than exact multilabel agreement, rare-label precision remains difficult, and raw language generation requires deterministic control. The design makes those limitations visible through per-finding evaluation, error analysis, Grad-CAM boundaries, raw-versus-guarded metrics and professional-review notices.

Future work should evaluate external datasets, calibrate probabilities under acquisition shift, compare stronger vision backbones, add expert-reviewed labels, introduce persistent encrypted storage, perform concurrency and load testing, evaluate subgroup performance, add model-drift monitoring, and conduct clinician-centered usability studies before any consideration of real-world deployment.

Final status: Notebook 9 reports SOLUTION READY FOR REPRODUCIBLE PACKAGING AND DEPLOYMENT. This means the educational prototype and its documented contracts are ready for submission and demonstration; it does not imply clinical readiness.

Appendix A. Complete Endpoint Catalogue

Method	Path	Purpose
GET	/health	Backend health and readiness without model inference
GET	/api/v1/model/info	Model, prompt, explainability and limitation lineage
GET	/api/v1/model/metrics	Persisted vision, language and guardrail metrics
GET	/api/v1/llmops/metrics	Operational request, latency and guardrail metrics
POST	/api/v1/image/classify	Validate and classify one chest X-ray
POST	/api/v1/image/analyze	Classify and return finding-specific visual evidence
POST	/api/v1/report/generate	Generate a grounded preliminary report
POST	/api/v1/explanation/generate	Generate a grounded plain-language explanation
POST	/api/v1/question/answer	Answer a question using stored prediction evidence
POST	/api/v1/follow-up/recommend	Generate controlled educational follow-up
POST	/api/v1/analyze-complete	Execute the complete image-to-language workflow
GET	/api/v1/predictions/{prediction_id}	Retrieve a stored prediction without repeated inference

Appendix B. Key Version and Artifact Traceability

Artifact/contract	Version or evidence
Computer-vision model	resnet18-chestmnist-v1
Language model	flan-t5-small-chestmnist-v1
Prompt registry	grounded-language-prompts-v1
Finding contract	chestmnist-finding-contract-v1
API contract	api-contract-v1
Readiness contract	fastapi-integration-readiness-v1
UI version	streamlit-interface-v1
Final registered files	49 checksum-validated artifacts
Final MLflow status	FINISHED
Final storage	8.71 GiB free; 4.00 GiB reserve preserved

References

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7. Hugging Face Transformers documentation. https://huggingface.co/docs/transformers/

8. FastAPI documentation. https://fastapi.tiangolo.com/

9. Streamlit documentation. https://docs.streamlit.io/

10. MLflow documentation. https://mlflow.org/docs/latest/

11. Project evidence: Notebooks 01-09, persisted API/UI validation artifacts and live screenshots supplied by Group 89.